SE 1105 - PROGRAMMING AND PROBLEM SOLVING I

Loops

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Loops

- Remember the function we had defined for calculating the passing grade of a student in our previous classes.

```
int grade(int lab, int mt, int fnl)
{
    return (lab + mt*4 + fnl*5) / 10;
}
```

- This function calculates only one student's grade. You can do it by calling this function in **main( )** once.

Loops

- What if you wanted to calculate the grades of 1000 students?

```
int grade(int lab, int mt, int fnl)
{
  return (lab + mt*4 + fnl*5) / 10;
}
void main()
{
 printf("Student#1's grade: %d\n",grade(20,40,60));
 printf("Student#2's grade: %d\n",grade(75,98,84));
 ...
 printf("Student#999's grade: %d\n",grade(67,56,71));
 printf("Student#1000's grade: %d\n",grade(90,95,100));
}
```

Loops

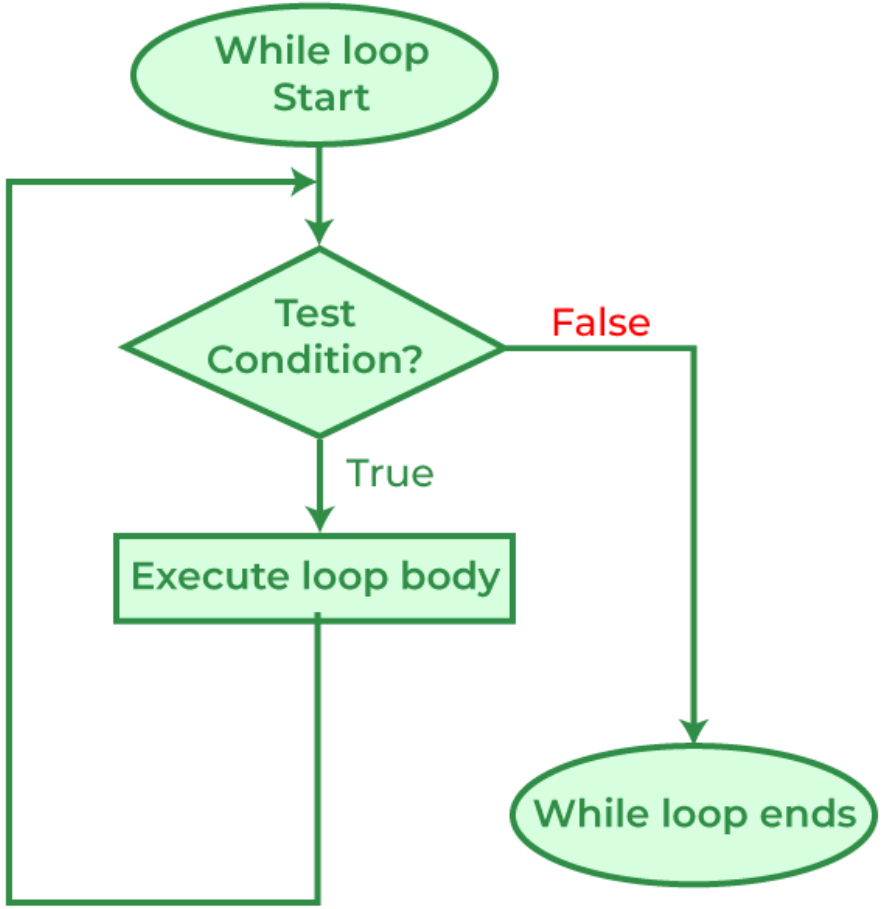
- Copying and pasting the same statement 1000 times is neither efficient nor correct. What if the number of repetitions also varies?
- Fortunately, we have a much better and correct way of doing repetitions in programming: a **loop**.
- A loop is a repetition statement that repeats a block of statements while a certain condition is true.
- This condition may be affiliated with the number of iterations or satisfying rational or logical statements.

Loops

- In C, we have three ways of constructing loops:

  1. while
  2. do-while
  3. for

While Loop




```
while ( conditions )
{

    ...statements...

}
```

Figure: https://www.geeksforgeeks.org/c-while-loop/?ref=header_outind

While Loop

Counting

```
1  #include<stdio.h>
2
3  void main()
4  {
5      int counter = 1
6      while ( counter <= 5 )
7      {
8          printf("%d\n",counter);
9          counter++;
10     }
11 }
```

Exercises

1. Write a function that takes two integers a and b, and returns the sum of all numbers between a and b.
2. Write a function that takes two integers a and b, and prints all the even numbers between a and b.
3. Write a function that takes two integers a and b, and returns a^b.
4. Write a function that takes an integer m, and returns $m!$
5. Write a function that takes a positive integer n as input and returns the closest integer to the square root of n (not bigger than n).

Do-While Loop

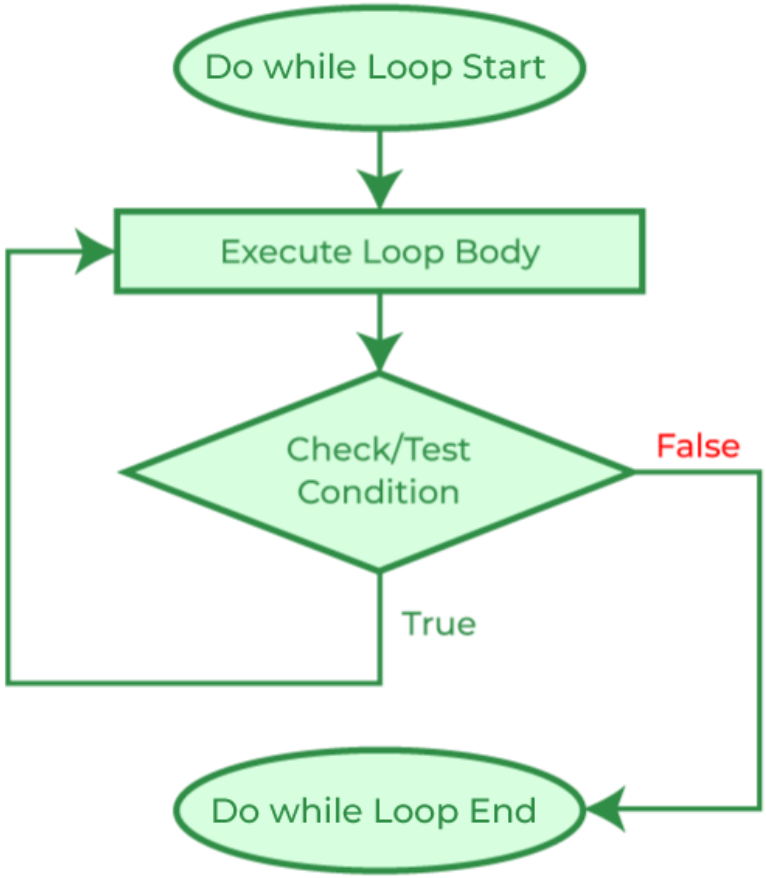


Figure: https://www.geeksforgeeks.org/c-do-while-loop/?ref=header_outind

```
do
{

    ...statements...

}while (conditions);
```

- Do-while loops execute their block of statements at least once!
- This is the only main difference between the **while** and **do-while** loops.

Do-While Loop

Counting

```
#include<stdio.h>

void main()
{
    int counter = 1
    do{
        printf("%d\n",counter);
        counter++;
    }while ( counter <= 5 );
}
```

For Loop

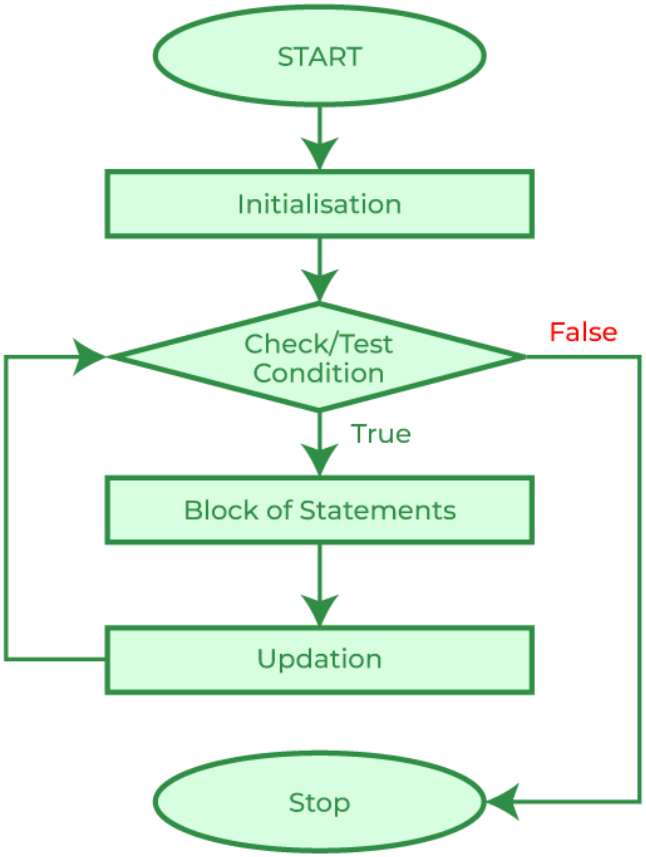


Figure: https://www.geeksforgeeks.org/c-for-loop/?ref=header_outind

```
for(initial;conditions;update)
{
    ...statements...
}
```

For Loop

Counting

```
1  #include<stdio.h>
2
3  void main()
4  {
5      int counter;
6      for(counter=1; counter<=5; counter++)
7      {
8          printf("%d\n",counter);
9      }
10 }
```

Exercises

1. *You can convert all the while loops in the previous exercises to for loops on your own.*

2. Write a function that prints the first n number of Fibonacci numbers.

$$F(0) = F(1) = 1,$$

F(n) = F(n-1) + F(n-2), for n $\geq$ 2

3. Write a function that takes an integer number and extracts its digits. For example, input:5078, output:5,0,7,8.

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Nested Loops

- Loops can be nested, as well.
- All three versions of loops can be used interchangeably as their inner loops.
- There is no limit to using the number of inner loops.
- Nested loops are usually used for controlling multiple variables individually or dealing with multi-dimensional arrays.

Exercises

1. Write a function that prints a square of edge-size n with "*"s.

2. How about these?

```
(A)
*
**
***
****
*****
******
*******
********
*********
**********

(B)
**********
*********
********
*******
******
*****
****
***
**
*

(C)
**********
 *********
  ********
   *******
    ******
     *****
      ****
       ***
        **
         *

(D)
         *
        **
       ***
      ****
     *****
    ******
   *******
  ********
 *********
**********
```

Figure: (Deitel and Deitel, 2022)

Exercises

3. Write a function that prints the Pythagorean Triples in the interval [1, 100]. Forexample, (3,4,5), (5,12,13), (8,15,17), etc.

4. Write a program that takes a positive number n and prints out the first n prime numbers.

References

Deitel, P. and Deitel, H. (2022). *C How to Program*. Pearson, 9th edition.